

1.

a.

$$\det A = a_{31}C_{31} + a_{35}C_{35} = 3C_{31} + (-1)C_{35}$$

$$M_{31} = -4[(-2)(-6) - 4(-21) + 6(-39)] + 8[4(-6) - 4(-12) + 6(-18)] + (-3)[4(-39) - (-2)(-18) + 4(15)]$$

$$= -4(-138) + 8(-84) + (-3)(-132) = 276$$

$$C_{31} = (-1)^{3+1}(276) = 276$$

$$M_{35} = 1[3(-24) - (-1)(-12) + 9(8)]$$

$$= 1(-12) = -12$$

$$C_{35} = (-1)^{3+5}(-12) = -12$$

$$\det A = 3(276) + (-1)(-12) = 828 + 12$$

$$\boxed{\det A = 840}$$

b.

$$M_{11} = (-1)[3(-24) - (-1)(-12) + 9(8)]$$

$$= (-1)(-12) = 12$$

$$M_{13} = -3[3(-9) - 9(-13) + 5(-12)]$$

$$= -3(30) = -90$$

$$M_{15} = -3[3(-24) - (-1)(-12) + 9(8)]$$

$$= -3(-12) = 36$$

$$M_{22} = -3[(-2)(-9) - 4(-20) + 6(-24)] + (-1)[1(-24) - (-2)(0) + 4(0)]$$

$$= -3(-46) + (-1)(-24) = 162$$

$$M_{23} = -3[4(-9) - 4(-13) + 6(-12)] + (-1)[1(-12) - 4(0) + 4(0)]$$

$$= -3(-56) + (-1)(-12) = 180$$

$$M_{24} = -3[4(-20) - (-2)(-13) + 6(8)] + (-1)[1(8) - 4(0) + (-2)(0)]$$

$$= -3(-58) + (-1)(8) = 166$$

$$M_{25} = -3[4(-24) - (-2)(-12) + 4(8)]$$

$$= -3(-88) = 264$$

$$M_{31} = -4[(-2)(-6) - 4(-21) + 6(-39)] + 8[4(-6) - 4(-12) + 6(-18)] + (-3)[4(-39) - (-2)(-18) + 4(15)]$$

$$= -4(-138) + 8(-84) + (-3)(-132) = 276$$

$$M_{32} = 1[(-1)(-9) - 9(-20) + 5(-24)]$$

$$= 1(69) = 69$$

$$M_{34} = 1[3(-20) - (-1)(-13) + 5(8)]$$

$$= 1(-33) = -33$$

$$M_{41} = -(-1)[4(-72) - (-2)(-36) + 4(28)]$$

$$= -(-1)(-248) = -248$$

$$M_{42} = 3[(-2)(-27) - 4(-37) + 6(-72)] - (-1)[1(-72) - (-2)(0) + 4(0)]$$

$$= 3(-230) - (-1)(-72) = -762$$

$$M_{43} = 3[4(-27) - 4(-29) + 6(-36)] - (-1)[1(-36) - 4(0) + 4(0)]$$

$$= 3(-208) - (-1)(-36) = -660$$

$$M_{44} = 3[4(-37) - (-2)(-29) + 6(28)] - (-1)[1(28) - 4(0) + (-2)(0)]$$

$$= 3(-38) - (-1)(28) = -86$$

$$M_{45} = 3[4(-72) - (-2)(-36) + 4(28)]$$

$$= 3(-248) = -744$$

$$M_{12} = -3[(-1)(-9) - 9(-20) + 5(-24)]$$

$$= -3(69) = -207$$

$$M_{14} = -3[3(-20) - (-1)(-13) + 5(8)]$$

$$= -3(-33) = 99$$

$$M_{21} = (-1)[4(-24) - (-2)(-12) + 4(8)]$$

$$= (-1)(-88) = 88$$

$$M_{33} = 1[3(-9) - 9(-13) + 5(-12)]$$

$$= 1(30) = 30$$

$$M_{35} = 1[3(-24) - (-1)(-12) + 9(8)]$$

$$= 1(-12) = -12$$

$$M_{51} = -(-1)[4(-39) - (-2)(-18) + 4(15)]$$

$$= -(-1)(-132) = -132$$

$$M_{52} = 3[(-2)(-6) - 4(-21) + 6(-39)] - (-1)[1(-39) - (-2)(0) + 4(0)]$$

$$= 3(-138) - (-1)(-39) = -453$$

$$M_{53} = 3[4(-6) - 4(-12) + 6(-18)] - (-1)[1(-18) - 4(0) + 4(0)]$$

$$= 3(-84) - (-1)(-18) = -270$$

$$\begin{aligned}
M_{54} &= 3[4(-21) - (-2)(-12) + 6(15)] - (-1)[1(15) - 4(0) + (-2)(0)] \\
&= 3(-18) - (-1)(15) = -39 \\
M_{55} &= 3[4(-39) - (-2)(-18) + 4(15)] \\
&= 3(-132) = -396
\end{aligned}$$

$$M = \begin{bmatrix} 12 & -207 & -90 & 99 & 36 \\ 88 & 162 & 180 & 166 & 264 \\ 276 & 69 & 30 & -33 & -12 \\ -248 & -762 & -660 & -86 & -744 \\ -132 & -453 & -270 & -39 & -396 \end{bmatrix}$$

c.

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$C = \begin{bmatrix} 12 & 207 & -90 & -99 & 36 \\ -88 & 162 & -180 & 166 & -264 \\ 276 & -69 & 30 & 33 & -12 \\ 248 & -762 & 660 & -86 & 744 \\ -132 & 453 & -270 & 39 & -396 \end{bmatrix}$$

d.

$$\text{adj } A = C^T = \begin{bmatrix} 12 & -88 & 276 & 248 & -132 \\ 207 & 162 & -69 & -762 & 453 \\ -90 & -180 & 30 & 660 & -270 \\ -99 & 166 & 33 & -86 & 39 \\ 36 & -264 & -12 & 744 & -396 \end{bmatrix}$$

e.

$$\begin{aligned}
R_3 &\rightarrow R_3 - 3R_1 && \rightarrow R_2 \rightarrow \frac{1}{3}R_2 \\
&= [0 \quad -12 \quad 6 \quad -12 \quad -19 \mid -3 \quad 0 \quad 1 \quad 0 \quad 0] = [0 \quad 1 \quad -\frac{1}{3} \quad 3 \quad \frac{5}{3} \mid 0 \quad \frac{1}{3} \quad 0 \quad 0 \quad 0] \\
&\rightarrow R_1 \rightarrow R_1 - 4R_2 && \rightarrow R_3 \rightarrow R_3 + 12R_2 \\
&= [1 \quad 0 \quad -\frac{2}{3} \quad -8 \quad -\frac{2}{3} \mid 1 \quad -\frac{4}{3} \quad 0 \quad 0 \quad 0] = [0 \quad 0 \quad 2 \quad 24 \quad 1 \mid -3 \quad 4 \quad 1 \quad 0 \quad 0] \\
&\rightarrow R_4 \rightarrow R_4 - 3R_2 && \rightarrow R_5 \rightarrow R_5 - 4R_2 \\
&= [0 \quad 0 \quad 5 \quad -6 \quad -4 \mid 0 \quad -1 \quad 0 \quad 1 \quad 0] = [0 \quad 0 \quad \frac{28}{3} \quad -12 \quad -\frac{29}{3} \mid 0 \quad -\frac{4}{3} \quad 0 \quad 0 \quad 1] \\
&\rightarrow R_3 \rightarrow \frac{1}{2}R_3 && \rightarrow R_1 \rightarrow R_1 + \frac{2}{3}R_3 && \rightarrow R_2 \rightarrow R_2 + \frac{1}{3}R_3 \\
&= [0 \quad 0 \quad 1 \quad 12 \quad \frac{1}{2} \mid -\frac{3}{2} \quad 2 \quad \frac{1}{2} \quad 0 \quad 0] = [1 \quad 0 \quad 0 \quad 0 \quad -\frac{1}{3} \mid 0 \quad 0 \quad \frac{1}{3} \quad 0 \quad 0] = [0 \quad 1 \quad 0 \quad 7 \quad \frac{11}{6} \mid -\frac{1}{2} \quad 1 \quad \frac{1}{6} \quad 0 \quad 0] \\
&\rightarrow R_4 \rightarrow R_4 - 5R_3 && \rightarrow R_5 \rightarrow R_5 - \frac{28}{3}R_3 \\
&= [0 \quad 0 \quad 0 \quad -66 \quad -\frac{13}{2} \mid \frac{15}{2} \quad -11 \quad -\frac{5}{2} \quad 1 \quad 0] = [0 \quad 0 \quad 0 \quad -124 \quad -\frac{43}{3} \mid 14 \quad -20 \quad -\frac{14}{3} \quad 0 \quad 1] \\
&\rightarrow R_4 \rightarrow -\frac{1}{66}R_4 && \rightarrow R_2 \rightarrow R_2 - 7R_4 \\
&= [0 \quad 0 \quad 0 \quad 1 \quad \frac{13}{132} \mid -\frac{5}{44} \quad \frac{1}{6} \quad \frac{5}{132} \quad -\frac{1}{66} \quad 0] = [0 \quad 1 \quad 0 \quad 0 \quad \frac{151}{132} \mid \frac{13}{44} \quad -\frac{1}{6} \quad -\frac{13}{132} \quad \frac{7}{66} \quad 0] \\
&\rightarrow R_3 \rightarrow R_3 - 12R_4 && \rightarrow R_5 \rightarrow R_5 + 124R_4 \\
&= [0 \quad 0 \quad 1 \quad 0 \quad -\frac{15}{22} \mid -\frac{3}{22} \quad 0 \quad \frac{1}{22} \quad \frac{2}{11} \quad 0] = [0 \quad 0 \quad 0 \quad 0 \quad -\frac{70}{33} \mid -\frac{1}{11} \quad \frac{2}{3} \quad \frac{1}{33} \quad -\frac{62}{33} \quad 1] \\
&\rightarrow R_5 \rightarrow -\frac{33}{70}R_5 && \rightarrow R_1 \rightarrow R_1 + \frac{1}{3}R_5 \\
&= [0 \quad 0 \quad 0 \quad 0 \quad 1 \mid \frac{3}{70} \quad -\frac{11}{35} \quad -\frac{1}{70} \quad \frac{31}{35} \quad -\frac{33}{70}] = [1 \quad 0 \quad 0 \quad 0 \quad 0 \mid \frac{1}{70} \quad -\frac{11}{105} \quad \frac{23}{70} \quad \frac{31}{105} \quad -\frac{11}{70}]
\end{aligned}$$

$$\begin{aligned}
& \rightarrow R_2 \rightarrow R_2 - \frac{151}{132}R_5 & \rightarrow R_3 \rightarrow R_3 + \frac{15}{22}R_5 \\
& = [0 \ 1 \ 0 \ 0 \ 0 \mid \frac{69}{280} \ \frac{27}{140} \ -\frac{23}{280} \ -\frac{127}{140} \ \frac{151}{280}] = [0 \ 0 \ 1 \ 0 \ 0 \mid -\frac{3}{28} \ -\frac{3}{14} \ \frac{1}{28} \ \frac{11}{14} \ -\frac{9}{28}] \\
& \rightarrow R_4 \rightarrow R_4 - \frac{13}{132}R_5 \\
& = [0 \ 0 \ 0 \ 1 \ 0 \mid -\frac{33}{280} \ \frac{83}{420} \ \frac{11}{280} \ -\frac{43}{420} \ \frac{13}{280}]
\end{aligned}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{70} & -\frac{11}{105} & \frac{23}{70} & \frac{31}{105} & -\frac{11}{70} \\ \frac{69}{280} & \frac{27}{140} & -\frac{23}{280} & -\frac{127}{140} & \frac{151}{280} \\ -\frac{3}{28} & -\frac{3}{14} & \frac{1}{28} & \frac{11}{14} & -\frac{9}{28} \\ -\frac{33}{280} & \frac{83}{420} & \frac{11}{280} & -\frac{43}{420} & \frac{13}{280} \\ \frac{3}{70} & -\frac{11}{35} & -\frac{1}{70} & \frac{31}{35} & -\frac{33}{70} \end{bmatrix} = \frac{1}{840} \text{adj } A$$

2.

a.

$$\begin{aligned}
& R_2 \rightarrow R_2 - 2R_1 & \rightarrow R_3 \rightarrow R_3 - R_1 \\
& = [(2 - 2(1)) \ (4 - 2(3)) \ (-1 - 2(4))] = [0 \ -2 \ -9] = [(1 - 1) \ (3 - 3) \ (-6 - 4)] = [0 \ 0 \ -10]
\end{aligned}$$

$$\text{REF} = \begin{bmatrix} 1 & 3 & 4 \\ 0 & -2 & -9 \\ 0 & 0 & -10 \end{bmatrix}$$

$$\det B = (1)(-2)(-10)$$

$$\boxed{\det B = 20}$$

b.

$$\det B = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$$

$$M_{11} = (4)(-6) - (-1)(3) = -21 \quad M_{12} = (2)(-6) - (-1)(1) = -11 \quad M_{13} = (2)(3) - (4)(1) = 2$$

$$\det B = 1(-21) - 3(-11) + 4(2) = -21 + 33 + 8$$

$$\boxed{\det B = 20}$$

c.

$$M_{11} = (4)(-6) - (-1)(3) = -21 \quad M_{12} = (2)(-6) - (-1)(1) = -11 \quad M_{13} = (2)(3) - (4)(1) = 2$$

$$M_{21} = (3)(-6) - (4)(3) = -30 \quad M_{22} = (1)(-6) - (4)(1) = -10 \quad M_{23} = (1)(3) - (3)(1) = 0$$

$$M_{31} = (3)(-1) - (4)(4) = -19 \quad M_{32} = (1)(-1) - (4)(2) = -9 \quad M_{33} = (1)(4) - (3)(2) = -2$$

$$C_{11} = +(-21) = -21, \quad C_{12} = -(-11) = 11, \quad C_{13} = +(2) = 2$$

$$C_{21} = -(-30) = 30, \quad C_{22} = +(-10) = -10, \quad C_{23} = -(0) = 0$$

$$C_{31} = +(-19) = -19, \quad C_{32} = -(-9) = 9, \quad C_{33} = +(-2) = -2$$

$$C = \begin{bmatrix} -21 & 11 & 2 \\ 30 & -10 & 0 \\ -19 & 9 & -2 \end{bmatrix} \rightarrow C^T$$

$$\boxed{\text{adj } B = \begin{bmatrix} -21 & 30 & -19 \\ 11 & -10 & 9 \\ 2 & 0 & -2 \end{bmatrix}}$$

d.

$$\begin{aligned}
& R_2 \rightarrow R_2 - 2R_1 \qquad \qquad \qquad \rightarrow R_3 \rightarrow R_3 - R_1 \qquad \qquad \qquad \rightarrow R_2 \rightarrow -\frac{1}{2}R_2 \qquad \qquad \rightarrow R_1 \rightarrow R_1 - 3R_2 \\
& = [0 \quad -2 \quad -9 \mid -2 \quad 1 \quad 0] = [0 \quad 0 \quad -10 \mid -1 \quad 0 \quad 1] = [0 \quad 1 \quad \frac{9}{2} \mid 1 \quad -\frac{1}{2} \quad 0] = [1 \quad 0 \quad -\frac{19}{2} \mid -2 \quad \frac{3}{2} \quad 0] \\
& \rightarrow R_3 \rightarrow -\frac{1}{10}R_3 \qquad \qquad \qquad \rightarrow R_1 \rightarrow R_1 + \frac{19}{2}R_3 \qquad \qquad \qquad \rightarrow R_2 \rightarrow R_2 - \frac{9}{2}R_3 \\
& = [0 \quad 0 \quad 1 \mid \frac{1}{10} \quad 0 \quad -\frac{1}{10}] = [1 \quad 0 \quad 0 \mid -\frac{21}{20} \quad \frac{3}{2} \quad -\frac{19}{20}] = [0 \quad 1 \quad 0 \mid \frac{11}{20} \quad -\frac{1}{2} \quad \frac{9}{20}]
\end{aligned}$$

$$B^{-1} = \begin{bmatrix} -\frac{21}{20} & \frac{3}{2} & -\frac{19}{20} \\ \frac{11}{20} & -\frac{1}{2} & \frac{9}{20} \\ \frac{1}{10} & 0 & -\frac{1}{10} \end{bmatrix}$$

$$\text{Verification: } B^{-1} = \frac{1}{\det B} \text{adj } B = \frac{1}{20} \begin{bmatrix} -21 & 30 & -19 \\ 11 & -10 & 9 \\ 2 & 0 & -2 \end{bmatrix} = \begin{bmatrix} -\frac{21}{20} & \frac{3}{2} & -\frac{19}{20} \\ \frac{11}{20} & -\frac{1}{2} & \frac{9}{20} \\ \frac{1}{10} & 0 & -\frac{1}{10} \end{bmatrix} \checkmark$$